



POLITECNICO
MILANO 1863

SPACECRAFT ATTITUDE DYNAMICS AND CONTROL

Lab 2.1 – Numerical Integration in Simulink

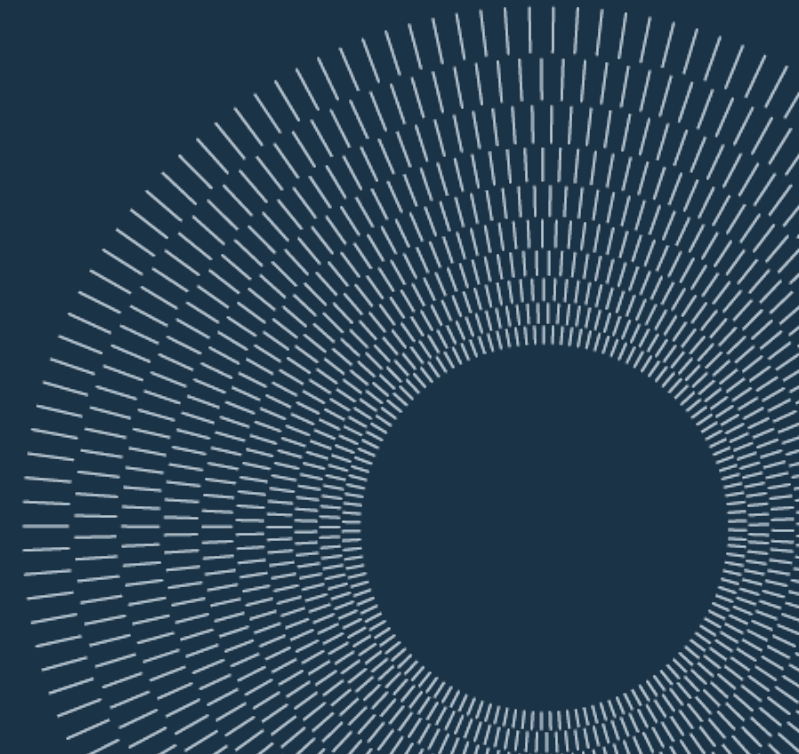
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A.Y. 2025-2026

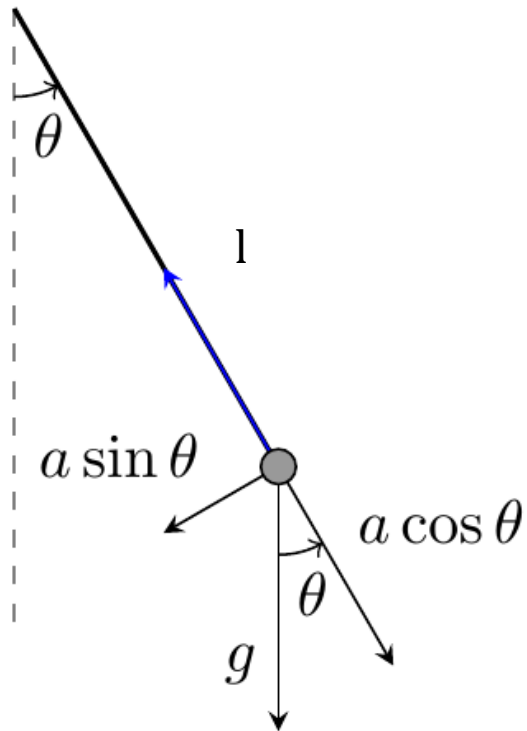
Examples



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Pendulum

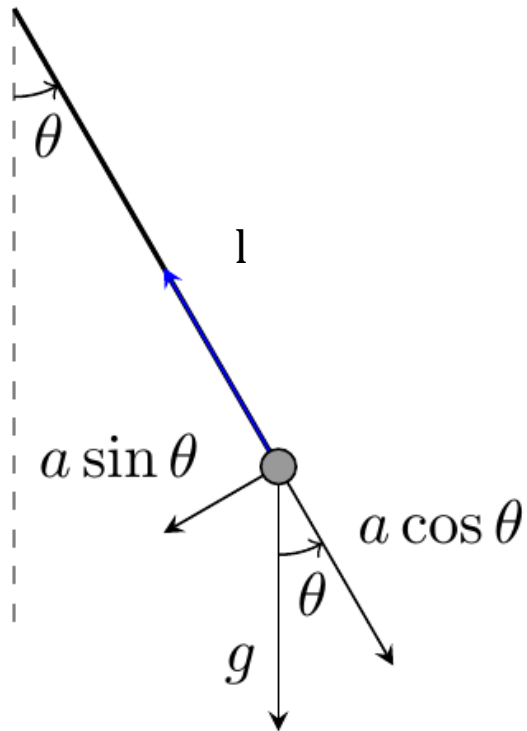


We want to simulate the motion of a pendulum under the effect of gravity.
We want to visualize both the angle θ and the angular velocity $\dot{\theta}$.

$$\ddot{\theta} = -\frac{g}{l} \sin \theta$$

$g = 9.81 \text{ m/s}^2$, $l = 5 \text{ m}$, $t_0 = 0 \text{ s}$, $t_f = 200 \text{ s}$
 θ_0 , $\dot{\theta}_0$ free, play with them.

Example: the simple pendulum



To simulate in Simulink higher order differential equations, it is useful to express them as system of first order equations (Cauchy problem), where:

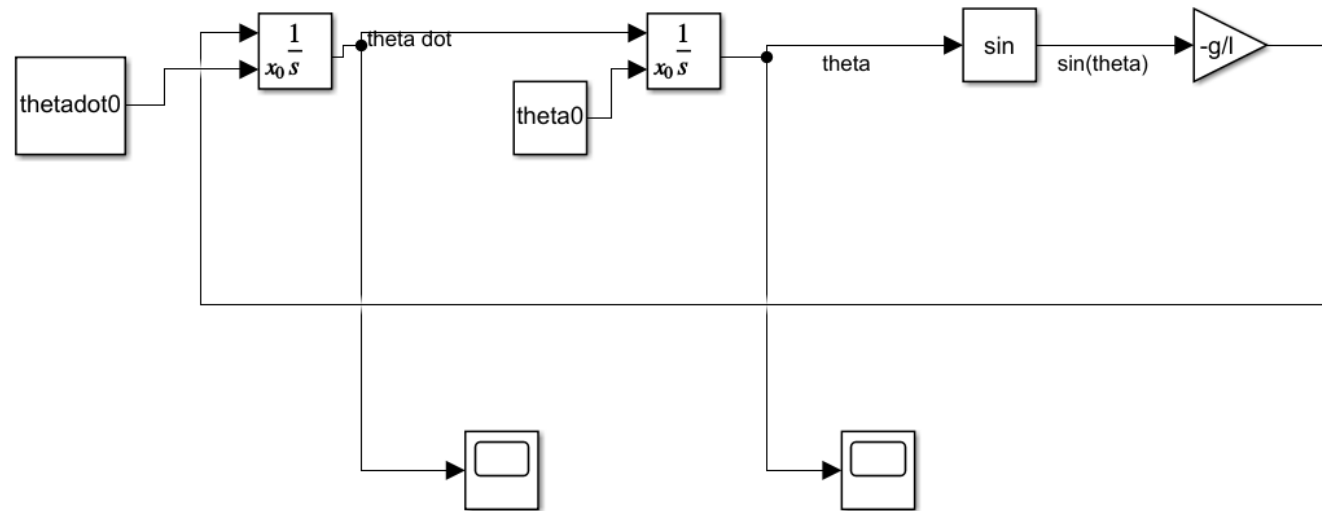
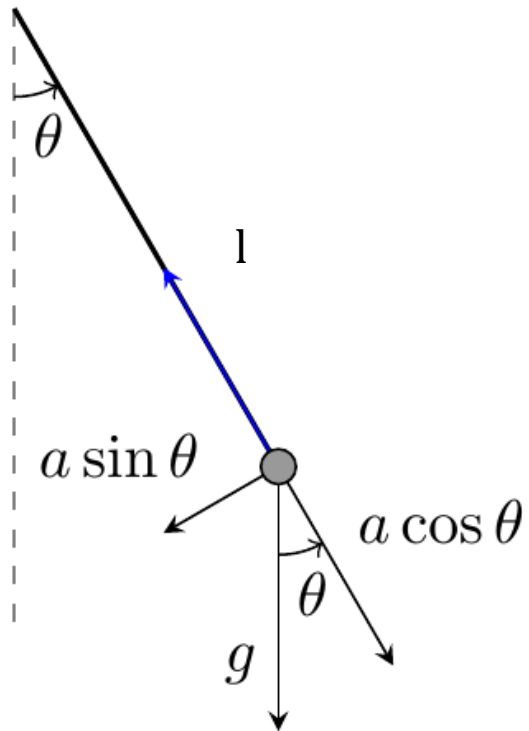
$$\ddot{\theta} = -\frac{g}{l} \sin \theta$$

Becomes

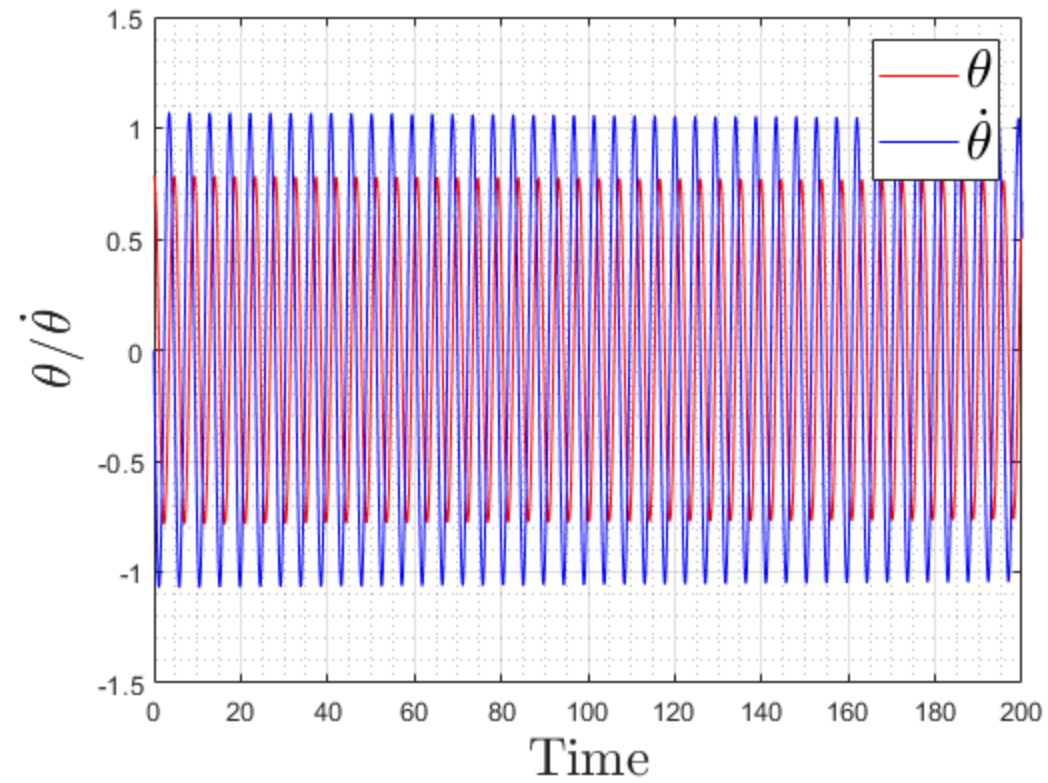
$$\begin{cases} \dot{x}_1(t) = x_2(t) \\ \dot{x}_2(t) = -\frac{g}{l} \sin(x_1(t)) \\ x_1(0) = \theta_0 \\ x_2(0) = \dot{\theta}_0 \end{cases}$$

with $\begin{cases} x_1(t) = \theta(t) \\ x_2(t) = \dot{\theta}(t) \end{cases}$

Pendulum



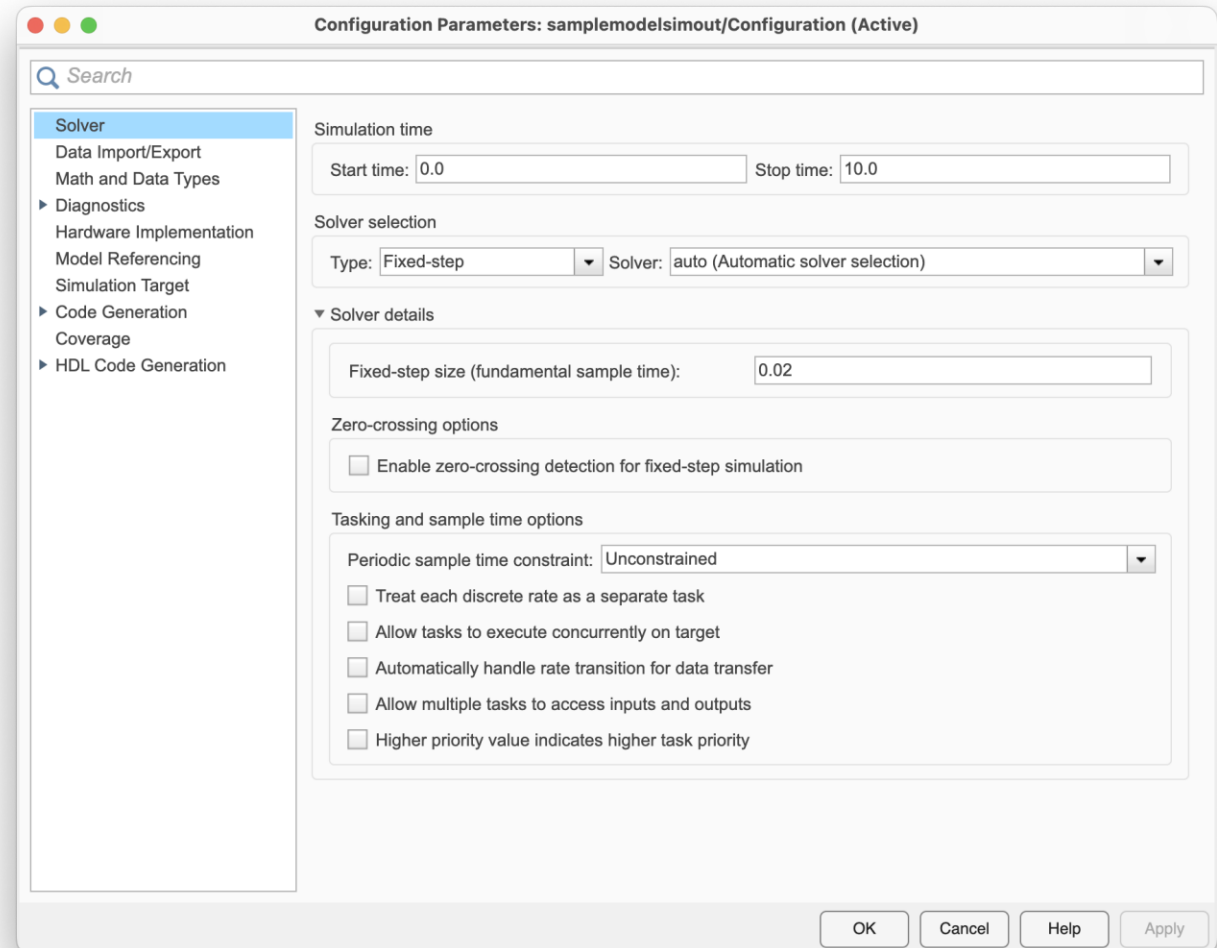
Pendulum



Solver setup in Simulink

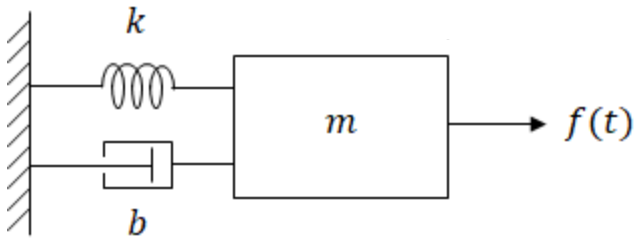
In Simulink, the solver settings can be accessed from the gear icon on the bottom-right:

Let's see them together.



Spring-Mass-Damper

We want to simulate the motion of a spring-mass-damper system under the effect of a force $f(t) = \sin(\omega t)$. We want to visualize both the position x and the velocity $v = \dot{x}$.



$$m\ddot{x} + b\dot{x} + kx = \sin(\omega t)$$

$$m = 2, b = 5, k = 6, \omega = 0.5$$

$$x_0 = 1, v_0 = 0$$

Useful resources



Simulink Onramp: a course on Simulink by the people who invented it